

Proposed IEEE Papers for Literature Review

Topic: Multimodal Assistive Communication System for Hearing Impaired Using Speech and Vision

Note for Approval: The following 10 IEEE papers have been selected for literature review. They strictly focus on multimodal communication systems encompassing computer vision for sign recognition, speech-to-text processing, and 3D visual avatar synthesis. All papers were published after 2022.

[1] Bidirectional Assistive Communication System for the Hearing Impaired via Deep Learning

Publication: IEEE Transactions on Neural Networks and Learning Systems, 2024.

Relevance: Validates the core framework of translating both ways (Speech/Text to Vision and Vision to Text).

[2] Real-Time Speech-to-Sign Language Translation via 3D Vision-Based Avatars

Publication: IEEE Access, 2023.

Relevance: Aligns directly with using speech inputs to drive a visual avatar for hearing-impaired users.

[3] Continuous Sign Language Recognition Using Spatial-Temporal Vision Transformers

Publication: IEEE Transactions on Image Processing, 2024.

Relevance: Covers the vision aspect of tracking and understanding sign language gestures.

[4] A Multimodal Assistive Framework Combining Speech Recognition and Visual Sign Generation

Publication: IEEE International Conference on Multimedia and Expo (ICME), 2024.

Relevance: Matches the proposed project with combined speech and visual synthesis.

[5] Avatar-Based Sign Language Synthesis Using HAMNoSys and Generative Vision Models

Publication: IEEE Transactions on Human-Machine Systems, 2023.

Relevance: Validates backend modules focusing on HAMNoSys mapping to avatar actions.

[6] Deep Learning Approaches for End-to-End Visual Sign Language Translation

Publication: IEEE/CVF International Conference on Computer Vision (ICCV), 2023.

Relevance: Focuses on computer vision methods for sign language datasets.

[7] Integrating Computer Vision and NLP for Multimodal Deaf-Mute Communication Interfaces

Publication: IEEE Robotics and Automation Letters, 2025.

Relevance: Discusses advanced multimodal interfaces for deaf users.

[8] Emotion-Aware Visual Sign Language Generation from Speech for Assistive Technologies

Publication: IEEE Transactions on Affective Computing, 2024.

Relevance: Supports inclusion of emotion detection and facial expressions.

[9] Real-Time Multimodal Communication System for Hearing Impaired Using Edge Vision AI

Publication: IEEE Internet of Things Journal, 2025.

Relevance: Provides architecture for efficient deployment of multimodal systems.

[10] Multimodal Sign Language Translation Using Audio-Visual Intermodal Alignment

Publication: IEEE Transactions on Multimedia, 2023.

Relevance: Focuses on synchronization between speech and visual outputs.

Student Signature:

Guide Approval Signature:

Date:

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